The Risk Journey of Gas Fumes

A 26-Session Worked Example Using WSCM-Lite

Maurice Codourey[1] and Emmanuel A. Gonzalez[2]

[1]codourey@proton.me
[2]emmgon@gmail.com

April 5, 2026

Abstract

This document provides a detailed narrative of a single weak signal—intermittent gas fumes near a loading dock—tracked across 26 cultivation sessions spanning 252 days. Every session is documented with the field observation, assessor scores, step-by-step calculations, and the resulting position on the cultivation field. The example uses the WSCM-Lite specification (biweekly cadence) and demonstrates the complete lifecycle of a risk signal from initial detection through escalation, emergency response, mitigation, and near-retirement. All computations can be reproduced with the accompanying Excel simulator.

Contents

1 Signal Definition

Signal name: Gas Fumes (Loading Dock B)

Scope: Intermittent detection of gas odor near loading dock B and adjacent storage corridor at an industrial facility.

Cadence: Biweekly. Gap boundaries: Early $\leq$ 10 days, Normal $\leq$ 21 days, Missed 1 $\leq$ 42 days, Missed 2+ $>$ 42 days.

Background: Three workers independently noticed a faint chemical smell near loading dock B over the month preceding the first formal cultivation session. None of the three observations were logged in any reporting system. At the next scheduled cultivation session, one of the workers raised the observation. The facilitator registered it as a new signal.

2 Phase 1: Question Marks (Sessions 1–5)

The signal enters and spends its first five sessions in the Question Marks region ($x < 5$, $y < 5$). During this phase, the team is uncertain about both the severity and growth trajectory. The signal gathers evidence slowly as more workers report observations.

Session 1 — Day 0 (Entry)

Field observation: A shift worker reports a faint, intermittent chemical smell near loading dock B. The smell appeared briefly during the afternoon shift and dissipated within minutes. No visible source identified. No equipment alarms triggered. The worker mentioned that two colleagues had noticed something similar on separate occasions over the past month.
Occurrences this period: 3 (pre-registration detections over the prior month).
Assessors: 1 (the reporting worker).
NRS scores:

Assessor	NRS_x (Intensity)	NRS_y (Growth)
Worker A	1	1

Calculation:

$$x_\text{new} = 2.5 \times 1.0 = 2.50$$
$$y_\text{new} = 2.5 \times 1.0 = 2.50$$

This is the entry session. The entry constraint is satisfied (all NRS scores $\leq$ 1). The signal is placed directly at the computed coordinates.

$$x' = 2.50 \qquad y' = 2.50$$
$$d = \sqrt{2.50^2 + 2.50^2} = \sqrt{12.50} = 3.54$$
$$f = 3 \qquad S = \frac{3.54}{14.14} \times \ln(1+3) = 0.250 \times 1.386 = 0.35 \text{ (Low)}$$

Position: (2.50, 2.50) — **Question Marks**.
SMS: No ($d = 3.54 < 7.07$).
Team decision: Log the signal. No action required. Monitor at next session.

Session 2 — Day 14

Field observation: The same worker reports one additional detection during the past two weeks. The smell appeared briefly near the same dock area, again during the afternoon shift. No other workers present at the time. No change in operations or equipment near the dock.
Occurrences this period: 1 (total f: 4).
Assessors: 1 (same worker).
NRS scores:

Assessor	$\mathbf{NRS_x}$	$\mathbf{NRS_y}$
Worker A	1	1

Gap classification: 14 days since Session 1. Biweekly cadence: $14 \leq 21 \rightarrow$ **Normal**. $w = 0.475$, decay $= 0.917$.
Calculation:

$$x_{\text{new}} = 2.50, \quad y_{\text{new}} = 2.50$$
$$c(1) = \min(1,\ 0.70 + 0.06 \times 1) = 0.76$$
$$w_{\text{eff}} = 0.475 \times 0.76 = 0.361$$

$$x' = 2.50 + 0.361 \times (2.50 - 2.50) = 2.50$$
$$y_{\text{decay}} = 2.50 \times 0.917 = 2.29$$
$$y' = \max(0.50,\ 2.29 + 0.361 \times (2.50 - 2.29)) = \max(0.50,\ 2.37) = 2.37$$
$$d = \sqrt{2.50^2 + 2.37^2} = 3.44$$
$$S = \frac{3.44}{14.14} \times \ln(5) = 0.243 \times 1.609 = 0.39 \text{ (Low)}$$

Position: $(2.50,\ 2.37)$ — **Question Marks**.
SMS: No.
Interpretation: The y coordinate decayed from 2.50 to 2.37. With no new evidence of escalation, growth urgency naturally diminishes. The signal remains stable.
Team decision: Continue monitoring. Ask dock workers to note the time and weather conditions if the smell recurs.

Session 3 — Day 28

Field observation: A second worker (Worker B) independently reports smelling chemical fumes near loading dock B. Worker B was unaware of Worker A's earlier reports. The smell was stronger than what Worker A described and persisted for approximately 15 minutes before dissipating. Worker B believes the growth potential is moderate because the smell seems to be getting more persistent.
Occurrences this period: 1 (total f: 5).
Assessors: 2 (Workers A and B).
NRS scores:

Assessor	$\mathbf{NRS_x}$	$\mathbf{NRS_y}$
Worker A	1	2
Worker B	1	2

Worker A maintains minimal intensity concern but agrees that the growth potential has increased. Worker B, having experienced a stronger smell, concurs with moderate growth.
Gap: 14 days → Normal. $w = 0.475$, decay $= 0.917$.
Calculation:

$$x_{\text{new}} = 2.5 \times \frac{1+1}{2} = 2.50, \quad y_{\text{new}} = 2.5 \times \frac{2+2}{2} = 5.00$$
$$c(2) = \min(1,\ 0.70 + 0.12) = 0.82, \quad w_{\text{eff}} = 0.475 \times 0.82 = 0.390$$
$$x' = 2.50 + 0.390 \times (2.50 - 2.50) = 2.50$$
$$y_{\text{decay}} = 2.37 \times 0.917 = 2.17$$
$$y' = \max(0.50,\ 2.17 + 0.390 \times (5.00 - 2.17)) = \max(0.50,\ 3.27) = 3.27$$
$$d = \sqrt{2.50^2 + 3.27^2} = 4.12, \quad S = \frac{4.12}{14.14} \times \ln(6) = 0.52 \text{ (Moderate)}$$

Position: $(2.50,\ 3.27)$ — **Question Marks**.
SMS: No.
Interpretation: The y coordinate jumped from 2.37 to 3.27. A second independent reporter with a higher growth score pulled y upward. The SSI crossed into Moderate for the first time.
Team decision: Ask maintenance to inspect gas lines and ventilation in the dock B area before the next session.

Session 4 — Day 42

Field observation: A third worker (Worker C) reports the smell. The fumes were detected on two separate days during the past two weeks, both times during warm afternoon conditions. Workers A and B also detected it again. The frequency is increasing. Maintenance inspected the area but found no obvious leak; they noted that the gas line routing runs beneath the dock floor.
Occurrences this period: 2 (total f: 7).
Assessors: 3 (Workers A, B, and C).
NRS scores:

Assessor	NRS_x	NRS_y
Worker A	2	2
Worker B	2	2
Worker C	2	2

All three assessors agree on moderate concern for both intensity and growth. The consensus reflects the increasing frequency and the unresolved source.
Gap: 14 days → Normal. $w = 0.475$, decay $= 0.917$.
Calculation:

$$x_{\text{new}} = 2.5 \times 2.0 = 5.00, \quad y_{\text{new}} = 2.5 \times 2.0 = 5.00$$
$$c(3) = 0.88, \quad w_{\text{eff}} = 0.475 \times 0.88 = 0.418$$
$$x' = 2.50 + 0.418 \times (5.00 - 2.50) = 2.50 + 1.05 = 3.55$$
$$y_{\text{decay}} = 3.27 \times 0.917 = 3.00$$
$$y' = \max(0.50,\ 3.00 + 0.418 \times (5.00 - 3.00)) = \max(0.50,\ 3.84) = 3.84$$
$$d = \sqrt{3.55^2 + 3.84^2} = 5.22, \quad S = \frac{5.22}{14.14} \times \ln(8) = 0.77 \text{ (Moderate)}$$

Position: $(3.55,\ 3.84)$ — **Question Marks**, but approaching the boundary.

SMS: No.
Team decision: Escalate the maintenance request. Request a formal gas detection survey of the sub-floor routing.

Session 5 — Day 49

Field observation: The gas detection survey has not yet been scheduled. Two workers report the smell again, and this time it was noticeably stronger—Worker A describes it as no longer "faint" but clearly present. Worker B was not on shift. Worker C confirms increased intensity. The smell persisted for over 30 minutes during the last occurrence.
Occurrences this period: 2 (total f: 9).
Assessors: 2 (Workers A and C).
NRS scores:

Assessor	$\mathrm{NRS_x}$	$\mathrm{NRS_y}$
Worker A	3	1
Worker C	3	1

Both assessors significantly increased their intensity score (from 2 to 3) but lowered growth potential to 1, reasoning that the smell seems to be constant rather than worsening.
Gap: 7 days → Early. $w = 0.281$, decay $= 0.957$.
Calculation:

$$\begin{aligned} x_{\text{new}} &= 2.5 \times 3.0 = 7.50, \quad y_{\text{new}} = 2.5 \times 1.0 = 2.50 \\ c(2) &= 0.82, \quad w_{\text{eff}} = 0.281 \times 0.82 = 0.230 \\ x' &= 3.55 + 0.230 \times (7.50 - 3.55) = 3.55 + 0.91 = 4.46 \\ y_{\text{decay}} &= 3.84 \times 0.957 = 3.67 \\ y' &= \max(0.50,\ 3.67 + 0.230 \times (2.50 - 3.67)) = \max(0.50,\ 3.40) = 3.40 \\ d &= \sqrt{4.46^2 + 3.40^2} = 5.61, \quad S = \frac{5.61}{14.14} \times \ln(10) = 0.91 \text{ (Moderate)} \end{aligned}$$

Position: $(4.46,\ 3.40)$ — **Question Marks**, near the $x = 5$ boundary.
SMS: No.
Interpretation: The x coordinate jumped from 3.55 to 4.46 as the team reported higher intensity. The y coordinate actually decreased because the new y score (2.50) was lower than the decayed previous position. The signal is being pulled rightward (more intense) but not yet upward (not yet escalating).
Team decision: Expedite the gas detection survey. Brief the shift supervisor.

3 Phase 2: Lit Fuses (Sessions 6–11)

The signal crosses $x = 5$ and enters Lit Fuses—high intensity, growth potential still below the midpoint. During this phase, the team is dealing with a clearly present hazard whose trajectory is uncertain. SMS escalation triggers at Session 6.

Session 6 — Day 63 **SMS TRIGGERED**

Field observation: The gas detection survey was finally conducted. Elevated methane readings were found in the sub-floor utility chase beneath dock B. The levels were below the lower explosive limit but well above background. Three workers independently con-

firmed that the smell is now a daily occurrence during warm afternoon shifts. The facility safety officer attended this session.
Occurrences this period: 3 (total f: 12).
Assessors: 3 (Workers A, B, and the safety officer).
NRS scores:

Assessor	$\mathrm{NRS_x}$	$\mathrm{NRS_y}$
Worker A	4	1
Worker B	4	1
Safety Officer	4	1

All three scored maximum intensity (confirmed gas readings) but low growth potential—the source is now identified and the readings are stable, not increasing.
Gap: 14 days → Normal. $w = 0.475$, decay $= 0.917$.
Calculation:

$$\begin{aligned}
x_{\text{new}} &= 2.5 \times 4.0 = 10.00, \quad y_{\text{new}} = 2.5 \times 1.0 = 2.50 \\
c(3) &= 0.88, \quad w_{\text{eff}} = 0.475 \times 0.88 = 0.418 \\
x' &= 4.46 + 0.418 \times (10.00 - 4.46) = 4.46 + 2.32 = 6.77 \\
y_{\text{decay}} &= 3.40 \times 0.917 = 3.12 \\
y' &= \max(0.50,\ 3.12 + 0.418 \times (2.50 - 3.12)) = \max(0.50,\ 2.86) = 2.86 \\
d &= \sqrt{6.77^2 + 2.86^2} = 7.35
\end{aligned}$$

$d = 7.35 \geq 7.07$ — **SMS escalation triggered.**
$S = \frac{7.35}{14.14} \times \ln(13) = 0.520 \times 2.565 = 1.33$ (Moderate).
Position: $(6.77,\ 2.86)$ — **Lit Fuses**.
SMS: YES — signal formally escalated to the Safety Management System.
Team decision: Initiate formal incident investigation. Restrict dock B access to essential personnel. Install continuous gas monitoring equipment. Report to facility management.

Session 7 — Day 70 SMS ACTIVE

Field observation: Continuous monitoring installed. Readings fluctuate—some days elevated, some days near background. The team suspects temperature-dependent gas migration through the sub-floor. Workers report the smell is less intense on cooler days.
Occurrences this period: 3 (total f: 15).
Assessors: 3. **NRS:** All score $(3, 2)$.
Gap: 7 days → Early. $w = 0.281$, decay $= 0.957$, $c(3) = 0.88$, $w_{\text{eff}} = 0.247$.

$$\begin{aligned}
x_{\text{new}} &= 7.50, \quad y_{\text{new}} = 5.00 \\
x' &= 6.77 + 0.247 \times (7.50 - 6.77) = 6.95 \\
y_{\text{decay}} &= 2.86 \times 0.957 = 2.74, \quad y' = 2.74 + 0.247 \times (5.00 - 2.74) = 3.30 \\
d &= 7.70, \quad S = 1.51 \text{ (Elevated)}
\end{aligned}$$

Position: $(6.95,\ 3.30)$ — **Lit Fuses**. SSI has crossed into Elevated.
Team decision: Continue monitoring. Schedule sub-floor inspection.

Session 8 — Day 77 **SMS ACTIVE**

Field observation: A hot day produced the highest gas reading yet. Workers describe the smell as "strong and unmistakable." However, the readings did not increase from the previous peak—the team views this as a recurring pattern, not an escalation.
Occurrences this period: 4 (total f: 19). **Assessors:** 3. **NRS:** All score $(4, 1)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.247$.

$$
\begin{aligned}
x_{\text{new}} &= 10.00, \quad y_{\text{new}} = 2.50 \\
x' &= 6.95 + 0.247 \times (10.00 - 6.95) = 7.71 \\
y_{\text{decay}} &= 3.30 \times 0.957 = 3.16, \quad y' = 3.16 + 0.247 \times (2.50 - 3.16) = 2.99 \\
d &= 8.27, \quad S = 1.75 \text{ (Elevated)}
\end{aligned}
$$

Position: $(7.71,\ 2.99)$ — **Lit Fuses**. Intensity climbing; growth suppressed.
Team decision: Accelerate sub-floor inspection timeline.

Session 9 — Day 91 **SMS ACTIVE**

Field observation: Sub-floor inspection completed. A corroded gas line joint was found but not yet repaired—parts on order. The corrosion explains the temperature-dependent leakage. Readings continue at the same elevated level. Workers report four more detections.
Occurrences this period: 4 (total f: 23). **Assessors:** 3. **NRS:** All score $(3, 2)$.
Gap: 14 days → Normal. $w_{\text{eff}} = 0.418$.

$$
\begin{aligned}
x' &= 7.71 + 0.418 \times (7.50 - 7.71) = 7.62 \\
y_{\text{decay}} &= 2.99 \times 0.917 = 2.74, \quad y' = 2.74 + 0.418 \times (5.00 - 2.74) = 3.69 \\
d &= 8.47, \quad S = 1.90 \text{ (Elevated)}
\end{aligned}
$$

Position: $(7.62,\ 3.69)$ — **Lit Fuses**. y creeping upward as the identified source remains unrepaired.
Team decision: Escalate repair priority. Temporary ventilation fan installed at dock B.

Session 10 — Day 98 **SMS ACTIVE**

Field observation: Repair parts still delayed. Four more occurrences recorded by monitoring equipment. Team consensus is that intensity remains high and growth potential is increasing because the corroded joint may worsen.
Occurrences this period: 4 (total f: 27). **Assessors:** 3. **NRS:** All score $(4, 2)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.247$.

$$
\begin{aligned}
x' &= 7.62 + 0.247 \times (10.00 - 7.62) = 8.21 \\
y_{\text{decay}} &= 3.69 \times 0.957 = 3.53, \quad y' = 3.53 + 0.247 \times (5.00 - 3.53) = 3.89 \\
d &= 9.08, \quad S = 2.14 \text{ (Elevated)}
\end{aligned}
$$

Position: $(8.21,\ 3.89)$ — **Lit Fuses**, approaching the $y = 5$ boundary.
Team decision: Issue formal safety alert. Restrict dock B to supervised access only.

Session 11 — Day 105 **SMS ACTIVE**

Field observation: A fourth assessor (the maintenance supervisor) joins. The corroded joint has visibly deteriorated. Gas readings spiked to the highest level yet during a heat wave. Six new occurrences logged. All assessors agree: both intensity and growth potential are significant.
Occurrences this period: 6 (total f: 33). **Assessors:** 4. **NRS:** All score $(3, 3)$.
Gap: 7 days → Early. $c(4) = 0.94$, $w_{\text{eff}} = 0.264$.

$$x' = 8.21 + 0.264 \times (7.50 - 8.21) = 8.02$$
$$y_{\text{decay}} = 3.89 \times 0.957 = 3.72, \quad y' = 3.72 + 0.264 \times (7.50 - 3.72) = 4.72$$
$$d = 9.31, \quad S = 2.32 \text{ (Elevated)}$$

Position: $(8.02,\ 4.72)$ — **Lit Fuses**, but y is now at 4.72, just below the $y = 5$ boundary.
Team decision: Emergency procurement for replacement parts. Management briefing scheduled.

4 Phase 3: Owls (Sessions 12–20)

The signal crosses into Owls ($x \geq 5$, $y \geq 5$)—high on both axes. This is the crisis phase. The signal reaches peak distance and peak SSI during this period. Emergency response is activated, repairs are completed, and the signal begins its descent.

Session 12 — Day 112 **SMS ACTIVE — EMERGENCY**

Field observation: Emergency. The corroded joint partially separated during a temperature spike. Gas alarms activated across dock B and the adjacent storage corridor. Area evacuated. Emergency ventilation deployed. Seven new occurrences recorded. All four assessors score maximum on both axes.
Occurrences this period: 7 (total f: 40). **Assessors:** 4. **NRS:** All score $(4, 4)$.
Gap: 7 days → Early. $c(4) = 0.94$, $w_{\text{eff}} = 0.264$.

$$x' = 8.02 + 0.264 \times (10.00 - 8.02) = 8.54$$
$$y_{\text{decay}} = 4.72 \times 0.957 = 4.52, \quad y' = 4.52 + 0.264 \times (10.00 - 4.52) = 5.97$$
$$d = \sqrt{8.54^2 + 5.97^2} = 10.42, \quad S = 2.74 \text{ (Critical)}$$

Position: $(8.54,\ 5.97)$ — **Owls**. Signal crosses $y = 5$ for the first time.
SSI enters Critical for the first time.
Team decision: Full emergency response. Dock B shut down. Emergency repair crew dispatched.

Session 13 — Day 119 **SMS ACTIVE — PEAK**

Field observation: Emergency repairs in progress. The corroded section has been isolated with a temporary bypass. Gas readings dropping but still elevated. Full committee of five assessors (facility manager joins). Eight new occurrences from the emergency period. All score maximum concern.
Occurrences this period: 8 (total f: 48). **Assessors:** 5. **NRS:** All score $(4, 4)$.
Gap: 7 days → Early. $c(5) = 1.00$, $w_{\text{eff}} = 0.281$.

$$x' = 8.54 + 0.281 \times (10.00 - 8.54) = 8.95$$
$$y' = 8.17 \times 0.957 + 0.281 \times (10.00 - 8.17 \times 0.957) = 6.92$$
$$d = 11.31, \quad S = 3.11 \text{ (Critical)}$$

Position: $(8.95,\ 6.92)$ — **Owls**. **Peak distance:** $d = 11.31$**.**
Team decision: Continue emergency repairs. Daily monitoring until readings normalize.

Session 14 — Day 133 **SMS ACTIVE**

Field observation: Temporary bypass holding. Permanent replacement pipe delivered. Gas readings declining but not yet at background. Four new occurrences during the two-week period. Full committee scores: intensity still maximum, but growth potential reduced—the situation is stabilizing.
Occurrences: 4 (total f: 52). **Assessors:** 5. **NRS:** All score $(4, 3)$.
Gap: 14 days → Normal. $w_{\text{eff}} = 0.475$.
$x' = 9.45$, $y' = 6.89$, $d = 11.70$, $S = 3.28$ (Critical).
Position: $(9.45,\ 6.89)$ — **Owls**. d peaks at 11.70.
Team decision: Schedule permanent repair for next maintenance window.

Session 15 — Day 140 **SMS ACTIVE**

Field observation: Permanent repair installed. Initial readings show significant improvement. However, residual gas detected in the sub-floor void—will take time to ventilate. Two new occurrences. Full committee: intensity dropping, growth potential still elevated due to residual gas.
Occurrences: 2 (f: 54). **Assessors:** 5. **NRS:** All score $(3, 4)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.281$.
$x' = 8.90$, $y' = 7.55$, $d = 11.67$, $S = 3.31$ (Critical). **Peak SSI.**
Position: $(8.90,\ 7.55)$ — **Owls**.

Session 16 — Day 147 **SMS ACTIVE**

Field observation: Gas readings continuing to decline. Sub-floor ventilation fans running. One new occurrence—a brief spike during fan maintenance. Four assessors remain (facility manager stepped back). Scores reflect declining concern on both axes.
Occurrences: 1 (f: 55). **Assessors:** 4. **NRS:** All score $(3, 3)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.264$.
$x' = 8.53$, $y' = 7.30$, $d = 11.23$, $S = 3.20$ (Critical).
Position: $(8.53,\ 7.30)$ — **Owls**. Both axes declining.

Session 17 — Day 154 **SMS ACTIVE**

Field observation: Source identified and repaired. Gas readings approaching background levels. One occurrence—a trace detection during a warm afternoon, but at much lower concentration. Workers report the smell is barely noticeable. Three assessors. Intensity scores drop to moderate; growth remains significant due to uncertainty about whether the repair will hold through summer.
Occurrences: 1 (f: 56). **Assessors:** 3. **NRS:** All score $(2, 3)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.247$.

$x' = 7.66$, $y' = 7.11$, $d = 10.45$, $S = 2.99$ (Critical).
Position: (7.66, 7.11) — **Owls**. x falling sharply post-repair.

Session 18 — Day 161 SMS ACTIVE

Field observation: One week post-repair. Readings at near-background. Only one trace detection, possibly residual. Workers report minimal concern. Intensity drops to minimal; growth potential remains due to summer heat ahead.
Occurrences: 1 (f: 57). **Assessors:** 3. **NRS:** All score $(1, 3)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.247$.
$x' = 6.38$, $y' = 6.98$, $d = 9.46$, $S = 2.72$ (Critical).
Position: (6.38, 6.98) — **Owls**. Intensity dropping rapidly.

Session 19 — Day 168 SMS ACTIVE

Field observation: No new detections by workers. Monitoring equipment shows background readings. Team agrees intensity is minimal but maintains growth concern—summer peak has not yet arrived.
Occurrences: 0 (f: 57). **Assessors:** 3. **NRS:** All score $(1, 3)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.247$.
$x' = 5.42$, $y' = 6.88$, $d = 8.76$, $S = 2.52$ (Critical).
Position: (5.42, 6.88) — **Owls**, but x approaching 5.

Session 20 — Day 182 SMS ACTIVE

Field observation: Two weeks since last session. One new occurrence—a trace detection during the hottest day of the month. Monitoring confirms the repair is holding, but the team scores growth potential high as a precaution given peak summer conditions.
Occurrences: 1 (f: 58). **Assessors:** 3. **NRS:** All score $(2, 4)$.
Gap: 14 days → Normal. $w_{\text{eff}} = 0.418$.
$x' = 5.25$, $y' = 7.85$, $d = 9.44$, $S = 2.72$ (Critical).
Position: (5.25, 7.85) — **Owls**, at the boundary.

5 Phase 4: Sleeping Cats (Sessions 21–24)

The signal crosses $x < 5$ while y remains above 5, entering Sleeping Cats—low current intensity but residual growth potential. The repair is holding, but the team maintains vigilance through summer.

Session 21 — Day 189 SMS ACTIVE

Field observation: No new detections. Monitoring shows clean readings. Workers are no longer reporting any smell. However, the team maintains high growth scores because a similar corrosion failure at another facility was reported in industry news.
Occurrences: 0 (f: 58). **Assessors:** 3. **NRS:** All score $(1, 4)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.247$.
$x' = 4.57$, $y' = 8.13$, $d = 9.32$, $S = 2.69$ (Critical).
Position: (4.57, 8.13) — **Sleeping Cats**. Signal crosses $x = 5$ downward.
Interpretation: Intensity has dropped below the midpoint, but growth potential is at its highest level in the signal's history. The team is concerned about systemic corrosion risk, not just this specific joint.

Team decision: Request corrosion survey of all gas line joints in the facility.

Session 22 — Day 196 SMS ACTIVE

Field observation: Facility-wide corrosion survey initiated. No new gas detections. Scores declining on both axes as confidence in the repair grows.
Occurrences: 0 (f: 58). **Assessors:** 3. **NRS:** All score $(1, 3)$.
Gap: 7 days → Early. $w_{\text{eff}} = 0.247$.
$x' = 4.06$, $y' = 7.71$, $d = 8.71$, $S = 2.51$ (Critical).
Position: (4.06, 7.71) — **Sleeping Cats**.

Session 23 — Day 210 SMS ACTIVE

Field observation: Corrosion survey completed. Two other joints flagged for preventive replacement but none showing active leakage. Growth potential dropping as systemic risk is being addressed. No occurrences in two weeks.
Occurrences: 0 (f: 58). **Assessors:** 3. **NRS:** All score $(1, 2)$.
Gap: 14 days → Normal. $w_{\text{eff}} = 0.418$.
$x' = 3.41$, $y' = 6.21$, $d = 7.08$, $S = 2.04$ (Elevated).
Position: (3.41, 6.21) — **Sleeping Cats**. d just barely above 7.07.
SSI drops from Critical to Elevated for the first time since Session 12.

Session 24 — Day 224

Field observation: No detections for four consecutive weeks. Monitoring equipment shows consistent background readings. Preventive replacements of the two flagged joints completed. Workers express confidence the issue is resolved. Two assessors remain (team returning to routine staffing).
Occurrences: 0 (f: 58). **Assessors:** 2. **NRS:** All score $(1, 2)$.
Gap: 14 days → Normal. $c(2) = 0.82$, $w_{\text{eff}} = 0.390$.
$x' = 3.05$, $y' = 5.42$, $d = 6.22$, $S = 1.79$ (Elevated).
Position: (3.05, 5.42) — **Sleeping Cats**.
SMS: No — $d = 6.22 < 7.07$. Signal drops below the SMS threshold for the first time since Session 6.
Team decision: Downgrade from active SMS tracking to routine cultivation monitoring. Maintain biweekly sessions for two more cycles to confirm stability.

6 Phase 5: Return to Question Marks (Sessions 25–26)

The signal crosses back below $y = 5$, returning to Question Marks. Both axes are declining. The signal is approaching retirement eligibility.

Session 25 — Day 238

Field observation: No detections. Background readings confirmed. Workers describe the issue as "resolved." Two assessors score minimal concern on both axes.
Occurrences: 0 (f: 58). **Assessors:** 2. **NRS:** All score $(1, 1)$.
Gap: 14 days → Normal. $c(2) = 0.82$, $w_{\text{eff}} = 0.390$.
$x' = 2.84$, $y' = 4.01$, $d = 4.91$, $S = 1.42$ (Moderate).
Position: (2.84, 4.01) — **Question Marks**. Signal has returned to its region of origin.
Team decision: One more session to confirm stability, then consider retirement.

Session 26 — Day 252 (Final)

Field observation: Eighth consecutive week with no detections. All monitoring readings at background. Preventive replacements completed across the facility. Workers confirm no residual smell. Two assessors maintain minimal scores.
Occurrences: 0 (f: 58). **Assessors:** 2. **NRS:** All score $(1, 1)$.
Gap: 14 days → Normal. $c(2) = 0.82$, $w_{\text{eff}} = 0.390$.
$x' = 2.71$, $y' = 3.22$, $d = 4.20$, $S = 1.21$ (Moderate).
Position: $(2.71,\ 3.22)$ — **Question Marks**.
SMS: No.
Retirement assessment: $d = 4.20 > 1.0$ — signal does not yet meet the retirement distance threshold. Although the team considers the issue resolved, the model reflects the accumulated history of a serious signal. The y coordinate remains at 3.22 (not near zero) because of the decay floor and the residual effect of months of elevated growth scores.
Team decision: Designate as a *retirement candidate*. Continue monitoring at monthly cadence. If d remains below 5.0 for three more sessions, formally retire the signal. The risk locus record is retained permanently.

7 Summary

S	Day	x'	y'	n	Region	Occ.	f	S	Key event
1	0	2.50	2.50	1	QM	3	3	0.35	Entry
2	14	2.50	2.37	1	QM	1	4	0.39	Same reporter
3	28	2.50	3.27	2	QM	1	5	0.52	Second reporter
4	42	3.55	3.84	3	QM	2	7	0.77	Third reporter
5	49	4.46	3.40	2	QM	2	9	0.91	Intensity spike
6	63	6.77	2.86	3	LF	3	12	1.33	**SMS triggered**
7	70	6.95	3.30	3	LF	3	15	1.51	Oscillating
8	77	7.71	2.99	3	LF	4	19	1.75	Peak hot day
9	91	7.62	3.69	3	LF	4	23	1.90	Source found
10	98	8.21	3.89	3	LF	4	27	2.14	Repair delayed
11	105	8.02	4.72	4	LF	6	33	2.32	Near boundary
12	112	8.54	5.97	4	OW	7	40	2.74	**Emergency**
13	119	8.95	6.92	5	OW	8	48	3.11	**Peak d**
14	133	9.45	6.89	5	OW	4	52	3.28	Bypass holding
15	140	8.90	7.55	5	OW	2	54	3.31	**Peak SSI**
16	147	8.53	7.30	4	OW	1	55	3.20	Repair installed
17	154	7.66	7.11	3	OW	1	56	2.99	Source fixed
18	161	6.38	6.98	3	OW	1	57	2.72	Readings dropping
19	168	5.42	6.88	3	OW	0	57	2.52	No detections
20	182	5.25	7.85	3	OW	1	58	2.72	Summer caution
21	189	4.57	8.13	3	SC	0	58	2.69	Enters Sl. Cats
22	196	4.06	7.71	3	SC	0	58	2.51	Survey started
23	210	3.41	6.21	3	SC	0	58	2.04	Survey done
24	224	3.05	5.42	2	SC	0	58	1.79	**Below SMS**
25	238	2.84	4.01	2	QM	0	58	1.42	Returns to QM
26	252	2.71	3.22	2	QM	0	58	1.21	Retirement cand.

Table 1: Complete 26-session Gas Fumes trajectory. QM = Question Marks, LF = Lit Fuses, OW = Owls, SC = Sleeping Cats.

7.1 Key Metrics

- **Total duration:** 252 days (36 weeks)
- **Region path:** Question Marks → Lit Fuses → Owls → Sleeping Cats → Question Marks
- **SMS active:** Sessions 6–23 (18 sessions, 147 days)
- **Peak distance:** $d = 11.70$ at Session 14

- **Peak SSI:** $S = 3.31$ at Session 15
- **Total real-world occurrences:** 58
- **Assessors:** ranged from 1 (entry) to 5 (peak crisis)